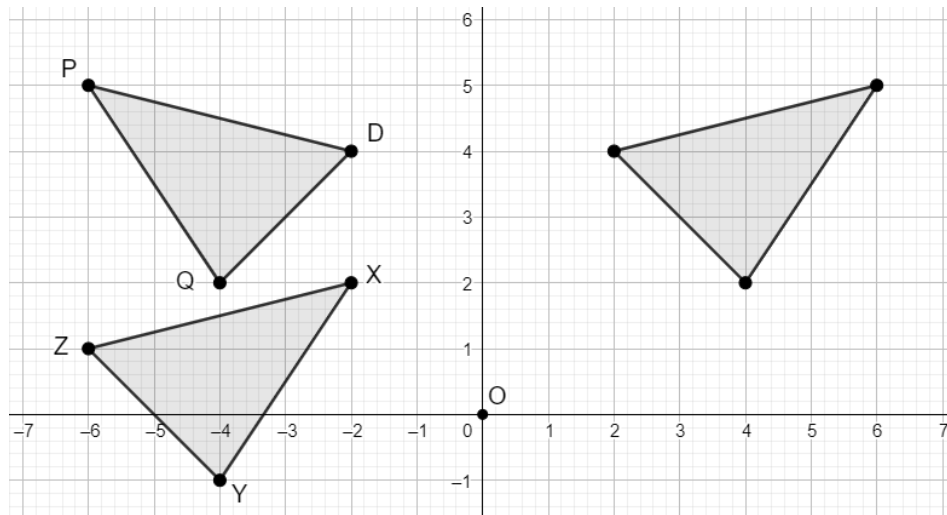


Geometry
Unit 6
Lesson 4 Practice

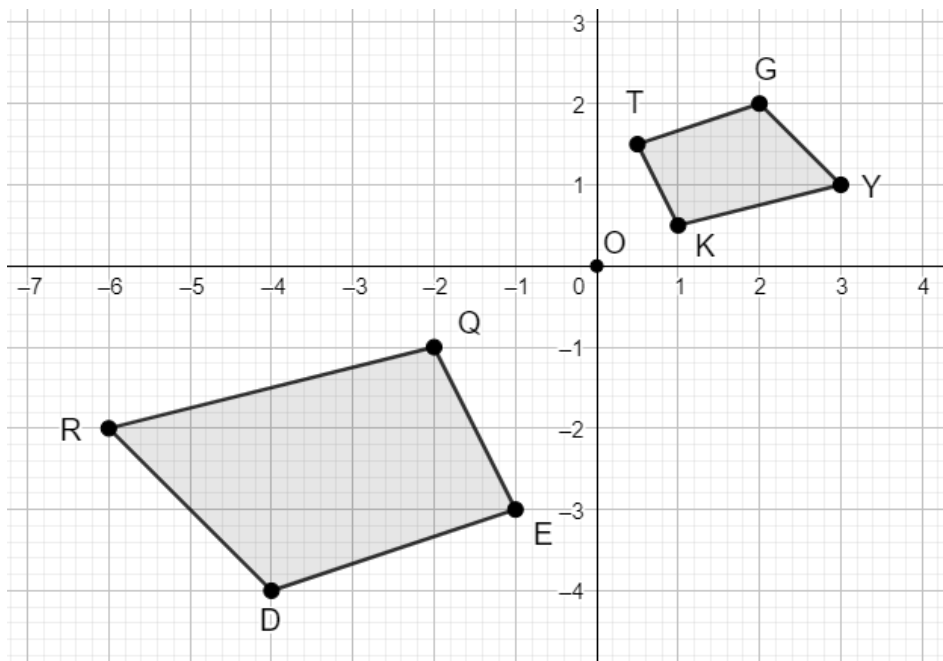
Name Answer Key
Date _____
Period _____

$\triangle XYZ$ is the result of a sequence of transformations of $\triangle PDQ$. The unlabelled triangle is the first transformation.



- Describe the first transformation of $\triangle PDQ$.
The first transformation is a reflection across the y-axis.
- Write the transformation algebraically.
 $(x, y) \rightarrow (-x, y)$
- Describe the second transformation that results in $\triangle XYZ$.
The second transformation is a translation 8 units left and 3 units down.
- Write the second transformation algebraically.
 $(x, y) \rightarrow (x - 8, y - 3)$
- Explain if the sequence of transformations preserves distance.
Yes, the sequence preserves distance because reflections and translations preserve distance.

KTGY is the result of a sequence of transformations of QEDR.



Ian and Natalie were determining the sequence of transformations, but each came up with a different sequence. Complete the table for the missing portions of their transformations.

			First Transformation	Second Transformation
6.	Ian's Sequence	Written Description	Rotate 180° about the origin.	$(x,y) \rightarrow (-x,-y)$
7.		Algebraic Description	Dilate using a scale factor of $\frac{1}{2}$ centered at the origin.	$(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$
8.	Natalie's Sequence	Written Description	Dilate using a scale factor of $\frac{1}{2}$ centered at the origin.	$(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$
9.		Algebraic Description	Rotate 180° about the origin.	$(x,y) \rightarrow (-x,-y)$

10. Considering both proposed sequences, you can conclude that

- ☐ Ian's sequence is

☐ Natalie's sequence is

☒ both sequences are

correct.